C-V CHARACTERISTICS OF A SOLAR CELL

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DATE OF EXPERIMENT - 18/02/2026 & 19/02/2026
DATE OF SUBMISSION -

Abstract

This experiment uses the CV characteristics of a solar cell to determine its internal parameters like doping density & built in potential. Using a trans-impedence amplifier circuit, the junction capacitance (C_{DOT}) is calculated as a function of applied reverse DC bias (0 - 1.5V) at a frequency of 5 kHz. The linea region of $\frac{1}{C^2}$ v/s V plot is analysed to calculate N_d & V_{bi}. The results confirm that as reverse bias increases, the depletion width expands, causing the capacitance to decrease.

Aim:-

To study how the junction capacitance varies with applied reverse bias and to estimate the doping density as well as the built-in potential.

Theory:-

A solar cell is similar to a p-n junction diode. In reverse bias of p-n junction, mobile charge carriers are swept away from the junction, leading to a depletion region of fixed donor & acceptor atoms. This depletion region acts like a parallel-plate capacitor. The depletion capacitance C is defined as rate of change of charge with applied DC voltage, $C = \frac{dQ}{dV_{DC}}$.

For a parallel-plate capacitor like depletion region,

$$C = \frac{\varepsilon_0 \varepsilon_s A}{x_d} \quad \text{—(I)}$$

where ε_0 = permittivity of free space
ε_s = dielectric constant of Si
A = junction area
x_d = depletion region width.

For a uniformly doped n-type region with donor concentration N_d, under reverse bias,

$$x_d = \left[\frac{2\varepsilon_0 \varepsilon_s (V_{bi} + V_{DC})}{q N_d}\right]^{1/2} \quad \text{—(II)}$$

V_{bi} = built-in potential
V_{DC} = applied reverse bias voltage
q = electronic charge
N_d = donor concentration.

put this in eq. (I),

$$C = \frac{\varepsilon_0 \varepsilon_s A}{\left[\frac{2\varepsilon_0 \varepsilon_s (V_{bi} + V_{DC})}{q N_d}\right]^{1/2}}$$

$$\Rightarrow \frac{1}{C^2} = \frac{2(V_{bi} + V_{DC})}{q \varepsilon_0 \varepsilon_s A^2 N_d} \quad \text{—(III)}$$

$\therefore$ $\frac{1}{C^2}$ & V_{DC} have a linear relation, (form) $y = mx + c$

From the linear relation

$$\frac{1}{C^2} = \frac{2}{q\varepsilon_0\varepsilon_s A^2 N_d} V_{DC} + \frac{2V_{bi}}{q\varepsilon_0\varepsilon_s A^2 N_d}$$

Let $\frac{1}{C^2} = mV_{DC} + c$.

The Doping concentration is calculated from the slope,

$$m = \frac{2}{q\varepsilon_0\varepsilon_s A^2 N_d} \Rightarrow \boxed{N_d = \frac{2}{q\varepsilon_0\varepsilon_s A^2 m}} \quad \text{-(IV)}$$

The intercept on X-axis @ $\frac{1}{C^2} = 0$ (extrapolation) we obtain $V_{DC} = -V_{bi}$

A small AC voltage is superimposed on the DC reverse bias:

$$\boxed{V_{DUT} = -R\left(\frac{V_{DC}}{R_2} + \frac{V_{AC}}{R_1}\right)} \quad \text{-(V)}$$

DUT : Device Under Test (Here, 'solar cell')

See figure - I, one end of solar cell is connected to the output of the summing circuit, while other is virtually grounded using negative feedback of OPAMP. Current through the capacitor is proportional to AC sinusoidal voltage. We use a transimpedence amplifier so that current flowing through the capacitor is converted into voltage. The trans-impedence amplifier generates a voltage output proportional to C_{DUT} & V_{DUT}. Magnitude of AC component of output voltage is :

$$\boxed{V_{out} = \omega V_{DUT} \frac{C_{DUT}}{C_F} \frac{1}{\sqrt{1 + \frac{1}{(\omega R_F C_F)^2}}}} \quad \text{-(VI)}$$

* Circuit Diagram :-

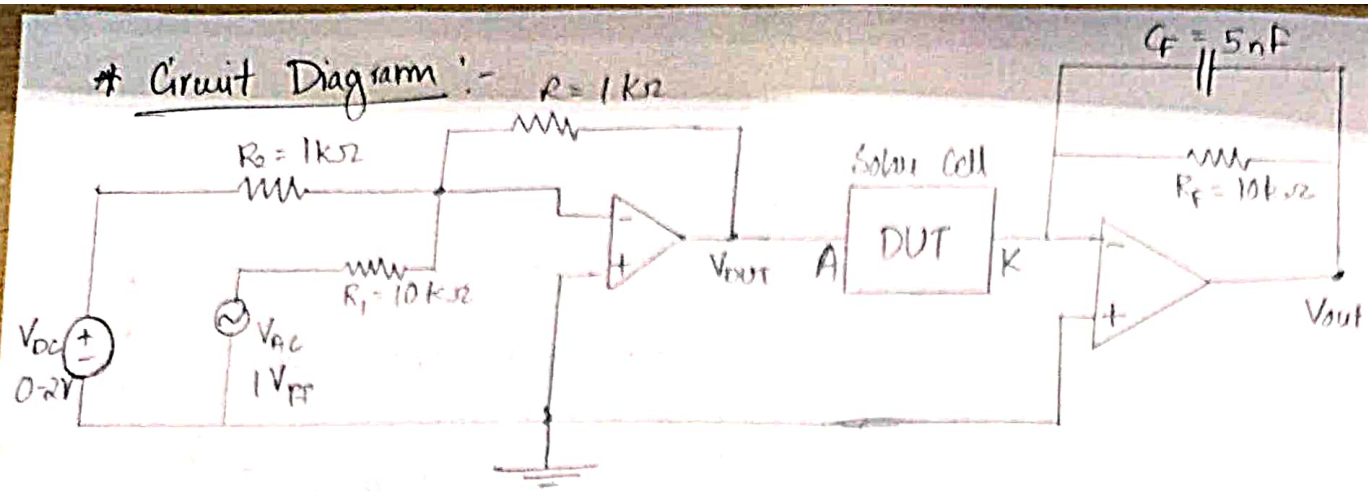


Fig I. Circuit for CV data, A - Anode of solar cell (DUT)
DUT = Device Under Test K - Cathode " " " "

Observations :-

S.no.	V_{DUT} (V)	V_{out} (AC) (V)	$V_{supplied}$ DC (V)
1.	0.0038	0.413	0
2.	-0.1306	0.401	0.1
3.	-0.1785	0.398	0.2
4.	-0.3470	0.385	0.3
5.	-0.3992	0.381	0.4
6.	-0.5067	0.374	0.5
7.	-0.618	0.367	0.6
8.	-0.689	0.363	0.7
9.	-0.809	0.356	0.8
10.	-0.933	0.349	0.9
11.	-1.022	0.345	1.0
12.	-1.091	0.342	1.1
13.	-1.191	0.338	1.2
14.	-1.324	0.331	1.3
15.	-1.401	0.327	1.4
16.	-1.545	0.321	1.5

NOTE:-
This data was taken on the first day, & when we performed the experiment on the second day, we observed very different readings & hence the data for "Dark" was taken again & this was "discarded".

18/01/26

Observations :-

Near Ambient Light

S.no.	V_{supply} (V)	V_{DUT} (V)	V_{out} (AC) (V)
1.	0	0.002	0.266
2.	0.1	-0.112	0.264
3.	0.2	-0.254	0.261
4.	0.3	-0.329	0.260
5.	0.4	-0.396	0.259
6.	0.5	-0.531	0.256
7.	0.6	-0.617	0.2565
8.	0.7	-0.748	0.254
9.	0.8	-0.800	0.253
10.	0.9	-0.914	0.252
11.	1.0	-1.064	0.251
~~12.~~	~~1.1~~	~~-1.209~~	~~0.250~~
13.	1.1	-1.134	0.251
14.	1.2	-1.252	0.250
15.	1.3	-1.320	0.250
16.	1.4	-1.438	0.249
17.	1.5	-1.529	0.248

Dark.

S.no.	V_{supply} (V)	V_{DUT} (V)	V_{out} (AC) (V)
1.	0	0.002	0.201
2.	0.1	-0.113	0.197
3.	0.2	-0.185	0.194
4.	0.3	-0.301	0.190
5.	0.4	-0.378	0.188
6.	0.5	-0.565	0.181
7.	0.6	-0.674	0.178
8.	0.7	~~-0.649~~ -0.739	0.176
9.	0.8	-0.831	0.174
10.	0.9	-0.961	0.172
11.	1.0	-1.050	0.168
12.	1.1	-1.142	0.166
13	1.2	-1.226	0.164
14.	1.3	-1.341	0.162
15.	1.4	-1.431	0.160
16.	1.5	-1.525	0.158

14/04/26

$$C_{DUT} = \frac{V_{out}}{V_{DUT}} C_F \sqrt{1 + \frac{1}{(\omega R_F C_F)^2}}$$

Calculations :-

A linear region was chosen from the $\frac{1}{C^2}$ v/s V_{DUT} plot and least square fitting was performed on those data points to obtain a linear fit equation

$$\frac{1}{C^2} = m V_{DUT} + C$$

where m = slope $\quad c$ = y-intercept

For N_d we use equation (IV), $\quad N_d = \frac{2}{q \varepsilon_0 \varepsilon_s A^2 m}$

For V_{bi}, (X - intercept).

$$0 = m V_{DUT} + C \Rightarrow V_{DUT} = V_{bi} = \frac{-|c|}{m}$$

(I) For Ambient light

From linear fit, we obtain,

$$\frac{1}{C^2} = m_1 V_{DUT} + C_1$$

where $m_1 = -1.22 \pm 0.04$

$C_1 = -0.81 \pm 0.05$

(plots & tables attached)

$$N_{d_1} = \frac{2}{q \varepsilon_0 \varepsilon_s A^2 m_1}$$

from eq. (III),

$$m = \frac{2}{q \varepsilon_0 \varepsilon_s A^2 N_d} \qquad c = \frac{2 V_{bi}}{q \varepsilon_0 \varepsilon_s A^2 N_d}$$

$A = 5\,cm \times 3.5\,cm = 17.5\,cm^2$

$\varepsilon_s = 11.7$

$N_d = 5.61 \times 10^{13}\ m^{-3}$

$V_{bi} = 0.66\,V$

(II) for Dark room:-

From linear fit plots & tables attached we obtain,

$$\frac{1}{C^2} = m_2 V_{DOT} + c_2 \quad \text{where} \quad m_2 = -3.10 \pm 0.12$$

$$c_2 = -2.16 \pm 0.15$$

$$N_{d_2} = \frac{2}{q \varepsilon_0 \varepsilon_s A^2 m_2} = 2.21 \times 10^{13} \ m^{-3}$$

$$V_{bi} \approx 0.70 \ V$$

Error Analysis :-

we have used the equation :

$$N_d = \frac{2}{q \varepsilon_0 \varepsilon_s A^2 m_2}$$

$$\Delta A = \sqrt{(0.1)^2 + (0.1)^2}$$

$$\Delta A = 0.1 \times \sqrt{2} \ cm^2$$

$$\Delta N_d = \sqrt{\left(\frac{\partial N_d}{\partial A} \Delta A\right)^2 + \left(\frac{\partial N_d}{\partial m} \cdot \Delta m\right)^2}$$

$$= \sqrt{\left(\frac{-2 \cdot 2}{q \varepsilon_0 \varepsilon_s A^2 m} \cdot \frac{\Delta A}{A}\right)^2 + \left(\frac{-1 \cdot 2}{q \varepsilon_0 \varepsilon_s A^2 m} \cdot \frac{\Delta m}{m}\right)^2}$$

$$\Delta N_d = N_d \sqrt{\left(\frac{2 \Delta A}{A}\right)^2 + \left(\frac{\Delta m}{m}\right)^2}$$

(I) for Ambient light

$$\Delta N_{d_1} = 5.61 \times 10^{13} \sqrt{\left(\frac{2 \times 0.1 \times \sqrt{2}}{17.5}\right)^2 + \left(\frac{0.04}{1.22}\right)^2}$$

$$\Delta N_{d_1} = 0.21 \times 10^{13} \ m^{-3}$$

(II) for Dark room

$$\Delta N_{d_2} = 2.21 \times 10^{13} \sqrt{\left(\frac{2 \times 0.1 \times \sqrt{2}}{17.5}\right)^2 + \left(\frac{0.12}{3.10}\right)^2}$$

$$N_{d_2} = 0.09 \times 10^{13} \ m^{-3}$$

Also, uncertainty in built-in potential

$$V_{bi} = \frac{c}{m}$$

$$\Delta V_{bi} = \sqrt{\left(\frac{\partial V_{bi}}{\partial c}\cdot\Delta c\right)^2 + \left(\frac{\partial V_{bi}}{\partial m}\cdot\Delta m\right)^2}$$

$$= \sqrt{\left(\frac{1}{m}\cdot\Delta c\right)^2 + \left(\frac{-c}{m^2}\cdot\Delta m\right)^2}$$

$$= \sqrt{\left(\frac{c}{m}\cdot\frac{\Delta c}{c}\right)^2 + \left(\frac{c}{m}\cdot\frac{\Delta m}{m}\right)^2}$$

$$\Delta V_{bi} = V_{bi}\sqrt{\left(\frac{\Delta c}{c}\right)^2 + \left(\frac{\Delta m}{m}\right)^2}$$

(I) for Ambient light

$m_1 = -1.22 \pm 0.04$

$c_1 = -0.81 \pm 0.05$

$$\Delta V_{bi_1} = 0.66\sqrt{\left(\frac{0.05}{0.81}\right)^2 + \left(\frac{0.04}{1.22}\right)^2}$$

$\Delta V_{bi_1} = 0.05\ V$

(II) for Dark room

$m_2 = -3.10 \pm 0.12$

$c_2 = -2.16 \pm 0.15$

$$\Delta V_{bi_2} = 0.70\sqrt{\left(\frac{0.15}{2.16}\right)^2 + \left(\frac{0.12}{3.10}\right)^2}$$

$\Delta V_{bi_2} = 0.06\ V$

Results :-

This experiment on C-V characteristics of a solar cell gave the following results :-

i) $1/C^2$ v/s V plot was analysed in the linear region for both cases (ambient light & dark room). The slope of the plot gave number density/doping concentration.

Case I :- Ambient light, $N_{d_1} = (5.61 \pm 0.21) \times 10^{13}\ m^{-3}$

Case II :- Dark room, $N_{d_2} = (2.21 \pm 0.09) \times 10^{13}\ m^{-3}$

ii) From the intercept, the built-in potential was calculated to be :-

Case I :- Ambient light, $V_{bi_1} = (0.66 \pm 0.05)V$

Case II :- Dark room, $V_{bi_2} = (0.70 \pm 0.06)V$

Conclusion :-

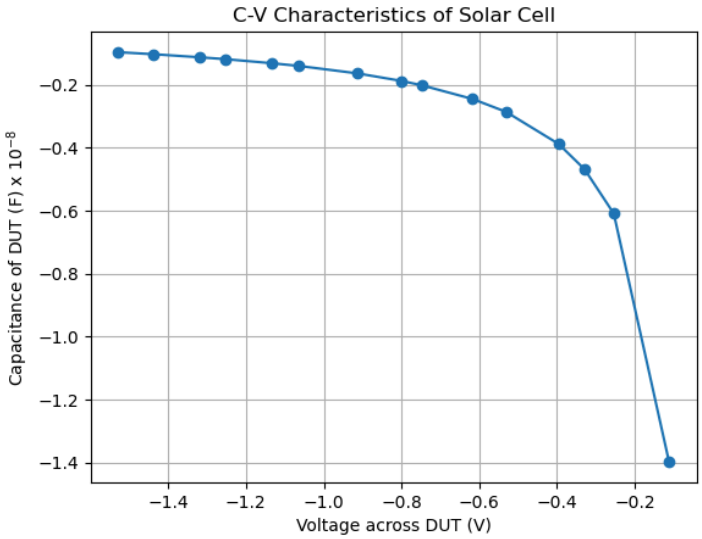
The experiment demonstrates the use of trans-impedence amplifier circuit to measure the voltage-dependent capacitance of a solar cell. A linear region was observed in the $\frac{1}{C^2}$ v/s V plot which validates the depletion approximation for a p-n junction under reverse bias. From the slope & intercept of the plot we determine the doping density & built-in potential of the solar cell. Therefore this method can be used to determine material parameters & properties of semi-conductor devices.

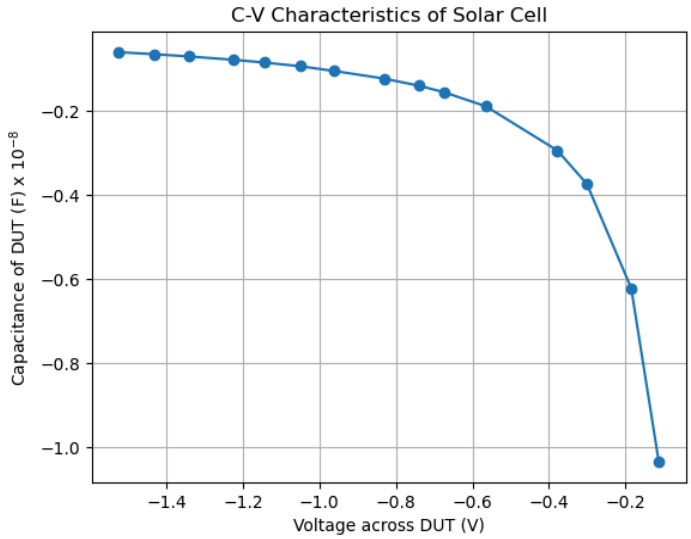
Source of Error :-

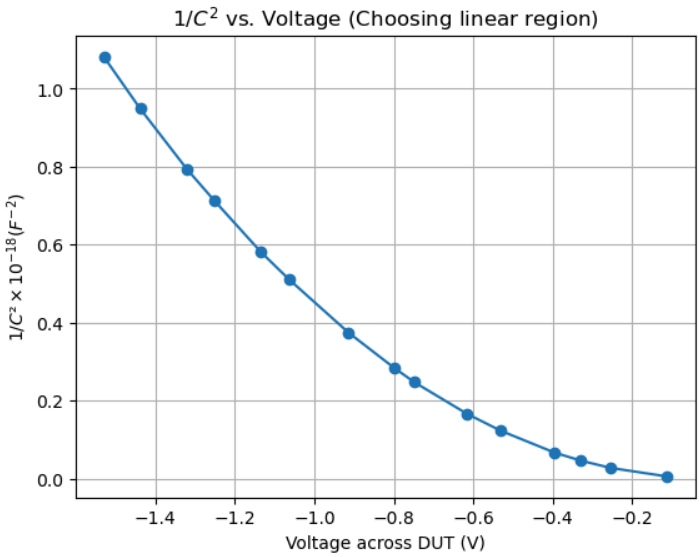
i) Precision of calculation C_{DOT} is limited by the sensitivity of the multimeters used to record small AC signal V_{out}.

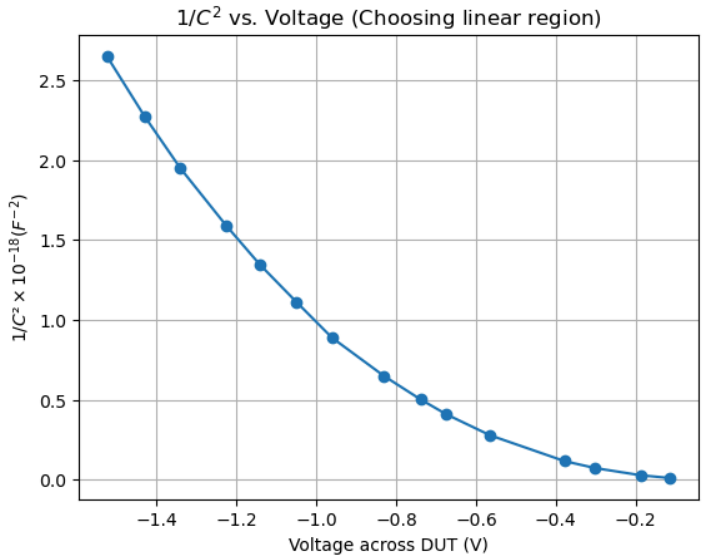
ii) Stray capacitance from breadboard & wires can affect the measurements.

iii) Light fluctuations near the setup can disturb the readings.



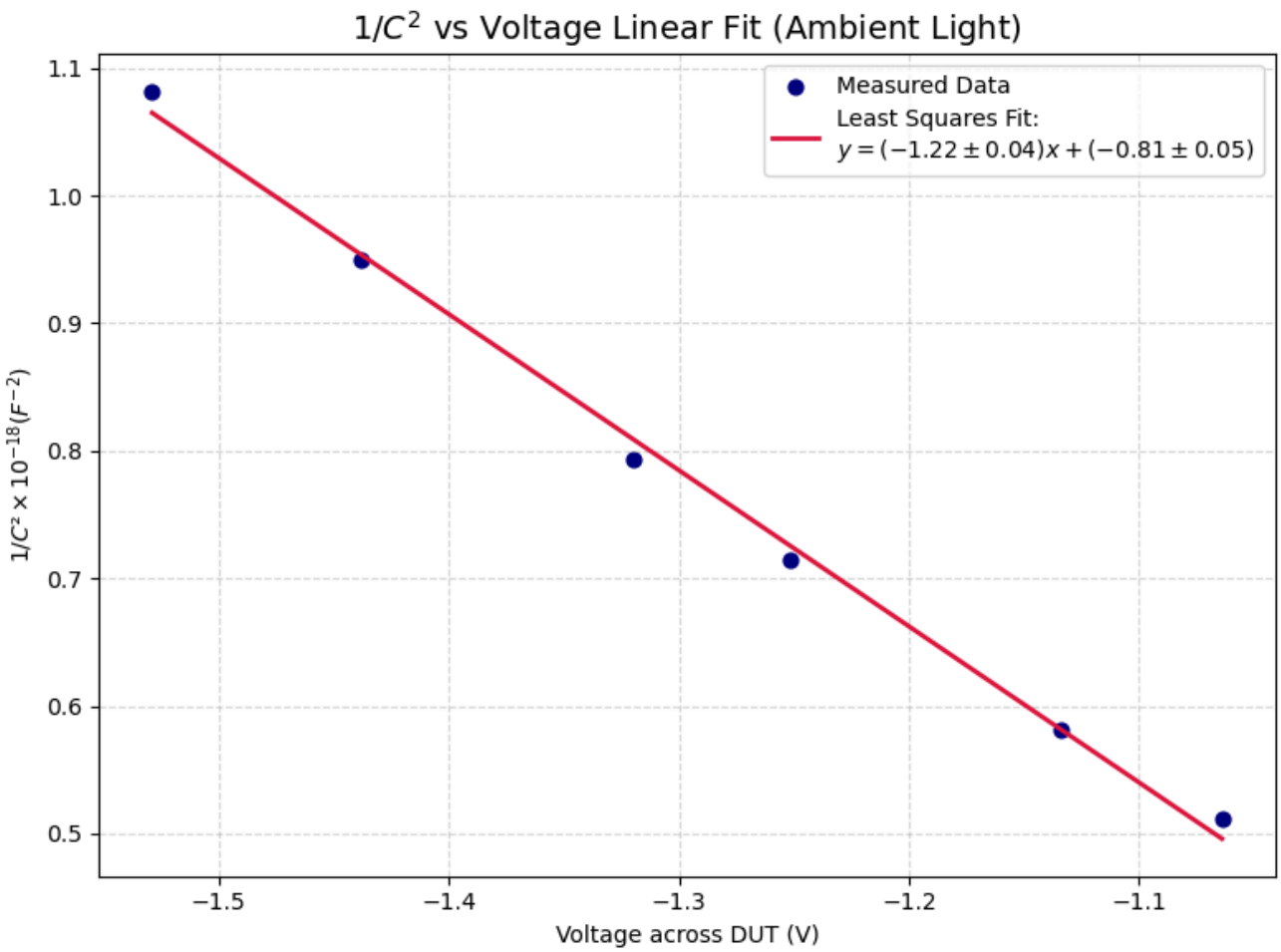






AMBIENT LIGHT

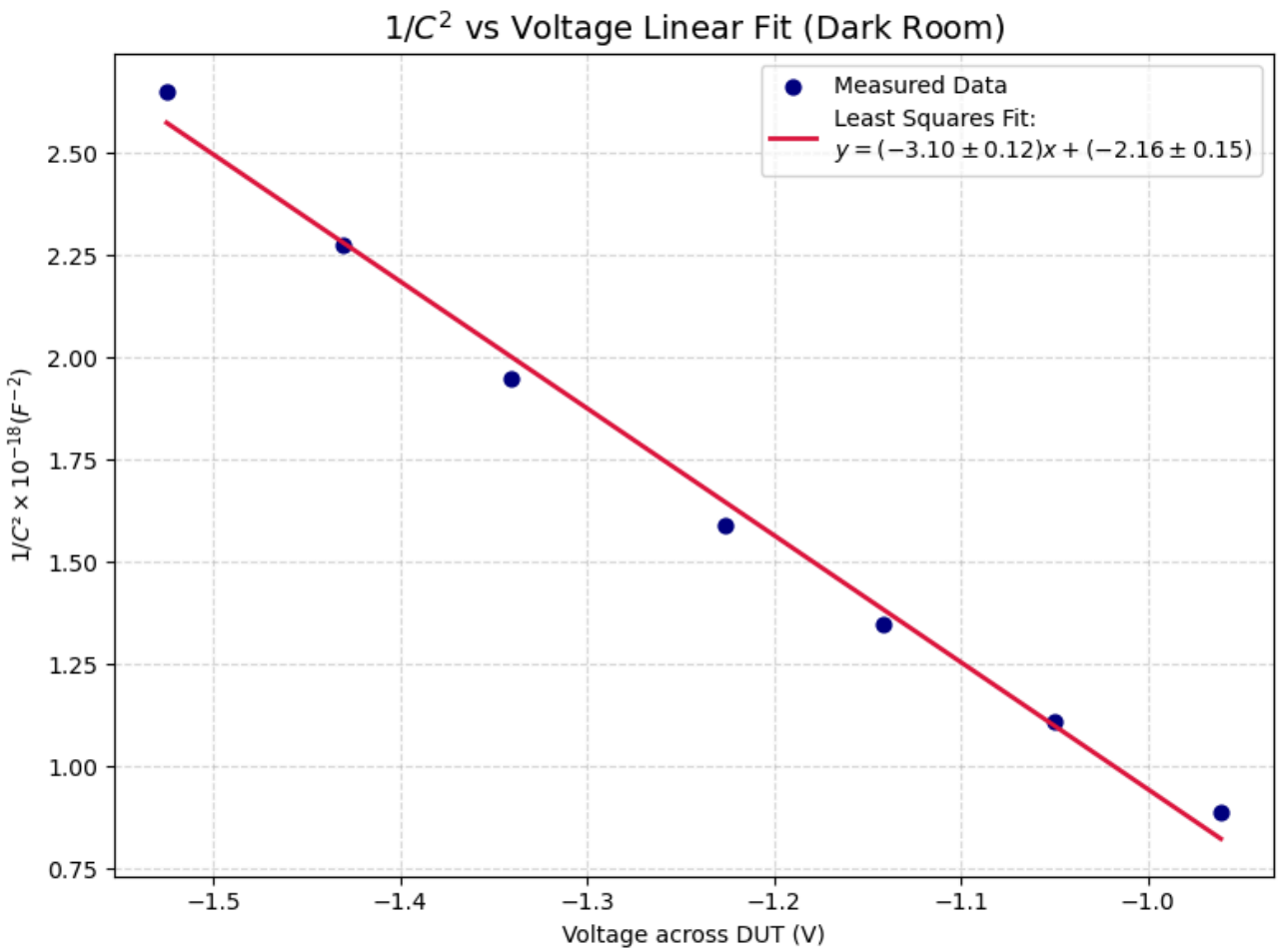
DARK ROOM



Input Data				
i	x	y	x^2	xy
1	-1.064	0.511	1.132	-0.544
2	-1.134	0.581	1.286	-0.659
3	-1.252	0.714	1.568	-0.894
4	-1.32	0.794	1.742	-1.047
5	-1.438	0.949	2.068	-1.365
6	-1.529	1.082	2.338	-1.654
Σ	-7.737	4.631	10.134	-6.164

Measured vs Fitted Values		
x	y (measured)	y (fit)
-1.064	0.511	0.496
-1.134	0.581	0.582
-1.252	0.714	0.726
-1.32	0.794	0.809
-1.438	0.949	0.954
-1.529	1.082	1.065

Fit Results & Statistical Errors		
Parameter	Value	Uncertainty
Slope (a_1)	-1.22	±0.04
Intercept (a_0)	-0.81	±0.05
σ_y (Std. Error)	0.02	
Δ (delta)	0.94	



Input Data				
i	x	y	x^2	xy
1	-0.961	0.889	0.924	-0.854
2	-1.05	1.112	1.102	-1.167
3	-1.142	1.347	1.304	-1.538
4	-1.226	1.591	1.503	-1.95
5	-1.341	1.95	1.798	-2.615
6	-1.431	2.277	2.048	-3.258
7	-1.525	2.652	2.326	-4.044
Σ	-8.676	11.817	11.005	-15.427

Measured vs Fitted Values		
x	y (measured)	y (fit)
-0.961	0.889	0.824
-1.05	1.112	1.1
-1.142	1.347	1.386
-1.226	1.591	1.647
-1.341	1.95	2.003
-1.431	2.277	2.283
-1.525	2.652	2.574

Fit Results & Statistical Errors		
Parameter	Value	Uncertainty
Slope (a_1)	-3.10	±0.12
Intercept (a_0)	-2.16	±0.15
σ_y (Std. Error)	0.06	
Δ (delta)	1.76	